

# Assignment: Real-Time Stream Processing with Azure Event Hubs & Stream Analytics

Course:

Deadline:

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## Objective

The goal of this assignment is to introduce you to **real-time data streaming and processing in Azure**.

You will:

- Create a free (no card required) “Azure for Students” account at: <https://azure.microsoft.com/en-us/free/students>
- Get the \$100 free credit
- Create an **Azure Event Hubs** instance
- Stream data into it using a provided **Python ATM transaction generator**
- Create multiple **Azure Stream Analytics jobs**
- Apply **different windowing techniques**
- Store all query results as **flat files in Azure Blob Storage**

A **bonus task** allows you to redirect output to an **Azure SQL Database** instead of Blob Storage.

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## Scenario

You are working for a bank that wants to analyze **ATM transactions in real time**.

Each ATM transaction event contains:

- Area where the ATM is located
- ATM identifier
- Account identifier
- Transaction type (withdrawal, deposit, transfer, balance inquiry)
- Transaction amount
- Event timestamp

You will process this stream to extract insights such as transaction volumes, high-value withdrawals, suspicious activity, and usage patterns.

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## Part 1: Environment Setup

Create the following in Azure:

- A **Resource Group**
  - An **Event Hubs Namespace**
  - An **Event Hub**
  - A **Storage Account**
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## Part 2: Python ATM Event Generator

You are provided with a Python script that generates **ATM transaction events** and sends them to Event Hubs.

### Requirements

- Modify the variables at the top of the script to match **your Event Hub**
- Run the script locally
- Verify that events are arriving in Event Hubs

The script continuously sends events with realistic transaction data at fixed intervals.

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## Part 3: Stream Analytics Jobs (Core Tasks)

You must create **five Stream Analytics jobs**, each with:

- Input: Event Hub
- Output: **Azure Blob Storage (JSON flat files)**
- Query: as described below

Each job must be **started and running**.

### **Task 1 – Transaction Volume per Area (Tumbling Window)**

#### **Description:**

Calculate the number of transactions and total transaction amount per area every **1 minute**.

**Window type:** Tumbling

### **Task 2 – High-Value Withdrawals (Hopping Window)**

**Description:**

Every minute, calculate how many withdrawals of €200 or more occurred in each area during the last 10 minutes.

**Window type:** Hopping

**Task 3 – Burst Activity per Account (Sliding Window)****Description:**

Detect accounts that perform 3 or more transactions within 30 seconds, indicating potential suspicious behavior.

**Window type:** Sliding

**Task 4 – Average Withdrawal Amount per Area (Tumbling Window)****Description:**

Every 5 minutes, calculate the average withdrawal amount per area, considering only withdrawal transactions.

**Window type:** Tumbling

**Task 5 – Transaction Type Distribution per Area (Simple Hopping Window)****Description:**

Every 5 minutes, calculate how many transactions of each type occurred in each area during the last 15 minutes.

**Window type:** Hopping

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**Output Requirements (Important)**

For **all five tasks**:

- Output must be written to **Azure Blob Storage**
- Use **JSON** format
- Each task must write to a **separate Blob container**
- Files must be automatically generated by Stream Analytics

Students should demonstrate that files are created and contain valid data.

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